THE RESEARCH PROCESS

NextInResearch

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IF IN A GROUP, PLEASE SIT WITH YOUR ASSIGNED GROUP

1. FOCUS + RESEARCH Q

What you have already done: identified a “field of interest” (computer science, neuroscience, chemistry, law, etc.)

Next step: Start with a general interest and ask: What is missing? What is misunderstood?

Effective topics are narrow enough to be manageable but broad enough to have scholarly relevance.

Begin your preliminary search:

- **Note overlapping and repeating issues**
- **Go down loopholes (easiest way to do this is to keep asking questions)**

Goal: Move from "I like climate change" to "How urban heat islands affect bird nesting in Baltimore."

TURN AND TALK

Turn to your neighbor and share your broad research interest. Your partner's job is to give you one constraint (a specific time, location, or population

- Example: "I like psychology."
- Constraint: "Focus on social media's impact on teenagers in the 2020s."

2. LITERATURE REVIEW

Your literature review will serve as your "specific dictionary", where you will:

- **Identify key theorists**
- **Define core terminology**
- **Spot methodological trends**

While also gaining insight on

- **What other scientists have done**
- **What skills you need to familiarize yourself with**

Look for:

- **Relevant articles**
- **Updated/recent articles**
- **Peer reviewed journals**

THIS PROCESS TAKES TIME!

STRUCTURE OF A RESEARCH PAPER

ABSTRACT

- Summary of problem, key findings, and significance (WRITE LAST)

INTRODUCTION

- Presents background, identifies knowledge gap, and states your hypothesis/ objective

MATERIALS + METHODS

- experimental design, procedures, and statistical tests.
- enough detail for another researcher to replicate the study

RESULTS

- State exactly what you found without interpretation.
- Well-labeled figures and tables.

DISCUSSION

- circle back to other authors + summarize meaning behind results

(sometimes) CONCLUSION

- sometimes merged with discussion

TURN AND TALK: HOW DO YOU READ A RESEARCH PAPER?

ANSWER: IT DEPENDS!

- ex: For a literature review, you may read a full paper, going abstract down to discussion
- ex 2: For finding methods, you may go from abstract directly to methods and results to see data analysis/statistical tests

2. HOW TO TAKE NOTES ON PAPERS

For every paper, ask yourself:
What did they study?
Why did they study it?
What methods did they use?
What did they find?
What limitations exist?
What questions remain?

The last question is the most important!

2. FINDING RESEARCH GAPS

DONT THINK:
"I need an impressive idea."

DO THINK:
What has yet to be studied or what limitation can I improve?

What conflicting results need explanation? →

Example:
Study A:
Exercise improves memory.
Study B:
Exercise has no effect.
Research Gap:
Does exercise influence memory differently depending on sleep quality?

TURN AND TALK

Read the abstract provided, and discuss the following questions with your group:

- What was the question?
- What did they find?
- What remains unanswered?

3. HYPOTHESIS

Note: hypothesis is NOT a guess, it is a PREDICTION supported by evidence

Aim to have around 3-5 supporting articles to justify your starting hypothesis (though more is better!)

If...then....

HOW TO BUILD:
Literature shows:
Exercise increases blood flow.
Memory depends on brain function.

Therefore:
Students who exercise before studying may perform better on memory tasks.

Hypothesis:
Students completing 30 minutes of aerobic exercise before studying will achieve higher memory test scores than students who do not exercise

3. STAYING ORGANIZED

EXPERIMENTAL DESIGN:
Can we trust the results?
Methods determine credibility, so take your time to understand the WHYs

IDENTIFYING VARIABLES
Example:
Does caffeine improve reaction time?
Independent Variable
Caffeine consumption
Dependent Variable
Reaction time
Control Variables:
Age
Sleep
Testing conditions
Time of day

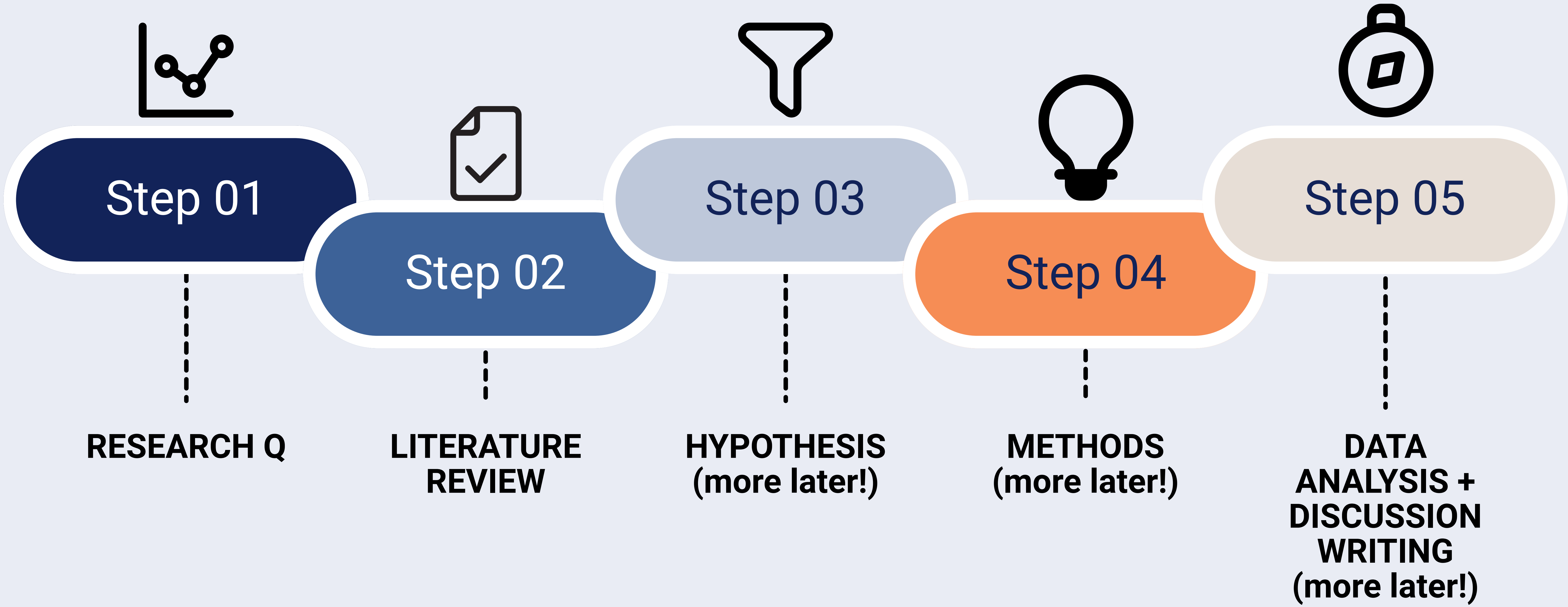
TURN AND TALK

"Three students drank coffee before a math test and scored low"

What can be improved?

- Missing controls
- Confounding variables

OVERVIEW OF PROCESS



GROUP ACTIVITY

TASK: Design a paper airplane that will TRAVEL THE FARTHEST

Materials:

- 3 sheets of paper
- Tape (ONLY for marking distance)
- You!

TWIST: You must write a research question and a hypothesis before running your experiment (the flight)

GROUP ACTIVITY

Was your hypothesis correct?
Did you make a control variable?
Independent and Dependent variables?
What went well? What didn't?

WINNING AIRPLANE...

YOUR NEXT STEPS

Stay tuned for an assignment *The Research PLAN*, where you will start finding articles and writing a rseearch quesiton with your mentor!

THANK YOU!